

Clinical Trial Design Proposal

STA 652

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1 Introduction and Background

1.1 Condition and Public Health Impact

Major depressive disorder (MDD) is a highly prevalent and disabling mental health condition characterized by persistent depressed mood, loss of interest, fatigue, impaired concentration, and changes in sleep and appetite. According to the National Institute of Mental Health, approximately 21 million U.S. adults experienced at least one major depressive episode in 2021, representing 8.3% of the adult population [National Institute of Mental Health, 2023]. Globally, the World Health Organization estimates that depression affects approximately 5% of adults worldwide [World Health Organization, 2023].

MDD is associated with substantial functional impairment, increased healthcare utilization, and economic burden due to lost productivity. Given its prevalence and impact, improving access to effective and scalable treatments remains a public health priority.

1.2 Current Standard Treatment

Cognitive behavioral therapy (CBT) is a well-established, evidence-based psychotherapy for depression and is commonly delivered in weekly face-to-face sessions over approximately 12 weeks. CBT focuses on identifying and restructuring maladaptive thoughts and behaviors contributing to depressive symptoms. Clinical guidelines support CBT as a first-line treatment for mild to moderate depression [American Psychological Association, 2019].

Although effective, traditional face-to-face CBT may be limited by cost, therapist availability, transportation barriers, and scheduling constraints. These barriers motivate evaluation of lower-cost, digitally delivered alternatives.

1.3 Outcome of Interest

The primary outcome for this study is depressive symptom severity measured using the Patient Health Questionnaire-9 (PHQ-9), a validated 9-item self-report instrument widely used in clinical trials and practice [Kroenke et al., 2001]. The PHQ-9 produces scores ranging from 0 to 27, with higher scores indicating more severe depressive symptoms.

The primary endpoint is the PHQ-9 score at 12 weeks following treatment initiation.

2 Study Objective and Hypotheses

The objective of this study is to evaluate whether a therapist-supported digital cognitive behavioral therapy program (dCBT), which is less expensive to deliver, is non-inferior to standard face-to-face CBT (FtF-CBT) in reducing depressive symptoms at 12 weeks.

Because the new intervention is expected to provide comparable clinical benefit while improving affordability and access, a non-inferiority design is appropriate.

Let μ_D denote the mean 12-week PHQ-9 score in the digital CBT group and μ_F denote the mean 12-week PHQ-9 score in the face-to-face CBT group. Define the treatment difference as:

$$\delta = \mu_D - \mu_F$$

where positive values indicate worse outcomes in the digital group.

The non-inferiority hypotheses are:

$$H_0 : \delta \geq \Delta \quad \text{vs} \quad H_1 : \delta < \Delta$$

where $\Delta = 2$ PHQ-9 points represents the maximum clinically acceptable difference.

A non-inferiority margin of 2 points was selected to represent a small difference unlikely to be clinically meaningful given the PHQ-9 scale range (0–27) and typical minimal clinically important differences reported in the literature.

3 Study Design

3.1 Trial Design

This study will be a two-arm, parallel-group, randomized controlled non-inferiority trial with 1:1 allocation. Participants will be followed for 12 weeks.

3.2 Target Population

The target population includes adults aged 18–65 years diagnosed with major depressive disorder and exhibiting at least moderate depressive symptoms.

Inclusion criteria:

- Age 18–65 years
- Clinical diagnosis of major depressive disorder
- Baseline PHQ-9 score ≥ 10
- Ability to provide informed consent

Exclusion criteria:

- Bipolar disorder or psychotic disorder
- Cognitive impairment limiting participation
- Immediate need for inpatient psychiatric care

3.3 Treatment Arms

Arm 1: Therapist-Supported Digital CBT (dCBT) Participants will complete structured CBT modules via a secure online platform and receive brief weekly therapist support (e.g., messaging or short telehealth check-ins). This model reduces therapist time and overall delivery cost.

Arm 2: Face-to-Face CBT (FtF-CBT) Participants will receive standard manualized CBT delivered in weekly 50–60 minute in-person sessions for 12 weeks.

3.4 Randomization and Confounding Control

Participants will be randomized 1:1 using a computer-generated allocation sequence. Randomization ensures balance of measured and unmeasured confounders across groups.

Stratified randomization may be implemented based on:

- Baseline PHQ-9 severity
- Current antidepressant medication use

The primary analysis will follow the intention-to-treat principle.

3.5 Primary Analysis Plan

The primary analysis will use an analysis of covariance (ANCOVA) model:

$$\text{PHQ9}_{12w} = \beta_0 + \beta_1 \cdot \text{Treatment} + \beta_2 \cdot \text{PHQ9}_{\text{baseline}} + \varepsilon$$

Adjusting for baseline PHQ-9 improves statistical precision by reducing residual variance and increasing power relative to an unadjusted comparison.

Non-inferiority will be declared if the upper bound of the two-sided 95% confidence interval for δ is strictly less than $\Delta = 2$, equivalent to a one-sided $\alpha = 0.025$ test.

4 Simulation-Based Power Analysis

A simulation-based power analysis was conducted under the following assumptions:

- Non-inferiority margin: $\Delta = 2$
- Standard deviation at 12 weeks: $\sigma = 6.9$
- True treatment difference: $\delta_{\text{true}} = 0$
- Baseline–follow-up correlation: $\rho = 0.5$
- One-sided $\alpha = 0.025$
- Target power: 0.90

The standardized non-inferiority margin is:

$$d = \frac{\Delta}{\sigma} = \frac{2}{6.9} \approx 0.29$$

For each candidate sample size, 2000 simulated trials were generated. In each simulation:

1. Baseline PHQ-9 values were simulated.
2. Follow-up PHQ-9 values were generated with correlation $\rho = 0.5$.
3. An ANCOVA model was fit.
4. The upper 95% confidence bound for δ was computed.
5. Non-inferiority was declared if the upper bound was less than 2.

Power was defined as the proportion of simulations concluding non-inferiority.

The smallest sample size achieving at least 90% power was 200 participants per group (400 total analyzable participants).

Assuming 20% attrition:

$$n_{\text{enroll}} = \frac{200}{0.80} = 250 \text{ per group}$$

Therefore, the total enrollment target is 500 participants.

5 Conclusion

This randomized non-inferiority trial is designed to evaluate whether therapist-supported digital CBT achieves clinically comparable outcomes to face-to-face CBT while improving accessibility and reducing cost. A simulation-based power analysis indicates that 500 total participants (250 per group) must be enrolled to achieve 90% power under the stated assumptions.

References

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